

Day 4 Tuning Report: Model Tuning, Regularization & Reproducible Pipelines

Adult Income Dataset — HistGradientBoosting selected as final model

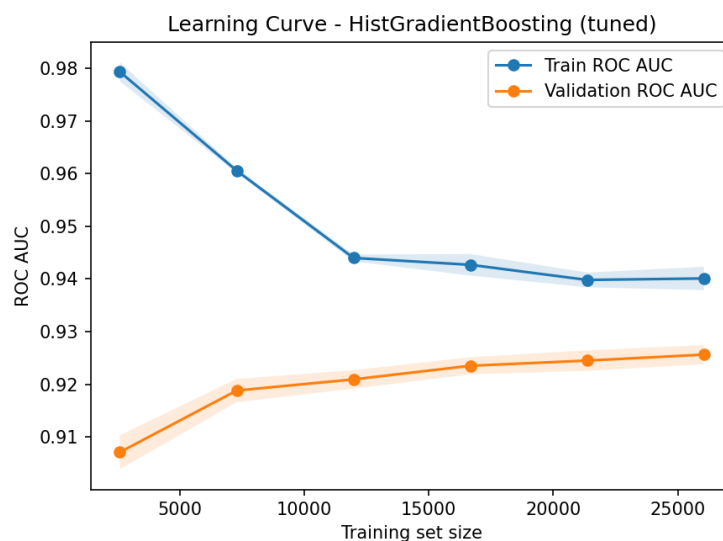
1. Hyperparameter Search

RandomizedSearchCV (StratifiedKFold, k=5, scoring=ROC AUC, n_jobs=-1) was run for all three candidates. Best configuration and 5-fold CV ROC AUC on the full training set after refitting:

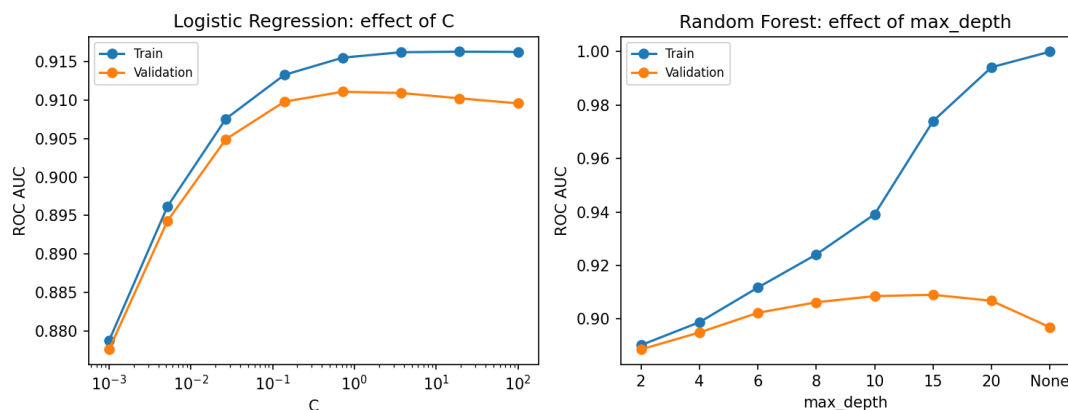
Model	Best hyperparameters	CV Accuracy	CV ROC AUC	CV F1
Logistic Regression	C=0.421, penalty=l1	0.858 ± 0.002	0.913 ± 0.001	0.674 ± 0.006
Random Forest	n_estimators=203, max_depth=14, min_samples_leaf=4, max_features=sqrt	0.859 ± 0.004	0.915 ± 0.002	0.654 ± 0.011
HistGradientBoosting	max_iter=160, learning_rate=0.177, max_depth=4, l2_regularization=0.961	0.872 ± 0.002	0.927 ± 0.002	0.710 ± 0.006

HistGradientBoosting wins on all three metrics and is selected as the final model to tune further, calibrate, and save.

2. Bias/Variance Diagnostics

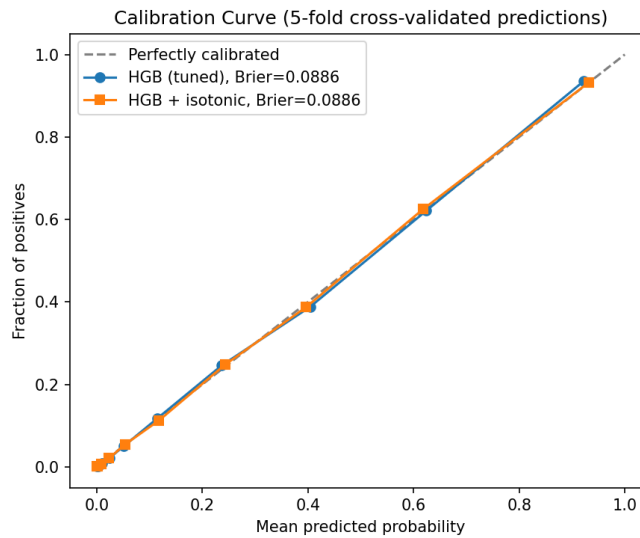


Learning curve (tuned HistGradientBoosting): the train/validation gap shrinks from ~ 0.07 at 10% of the data to ~ 0.014 at full training size — mild overfitting at small sizes that mostly resolves with more data. The validation curve is still gently rising at full size, so a bit more data would likely help further, but the current gap is small enough that the tuned hyperparameters are reasonable as-is.



Logistic Regression (left): validation ROC AUC peaks around $C \approx 0.7-1$, then flattens/dips as C grows further (less regularization) while train score keeps climbing — the fix is to keep C in the 0.1-1 range rather than defaulting to a large C. Random Forest (right): train score climbs toward ~ 1.0 as max_depth grows unconstrained, while validation peaks around depth $\approx 10-15$ then declines — the fix is capping max_depth rather than leaving it unlimited.

3. Calibration & Threshold Selection

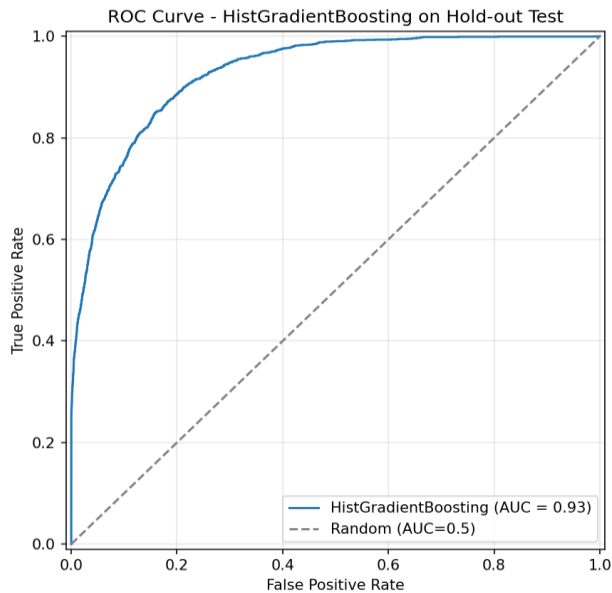


Brier score: 0.0886 uncalibrated vs. 0.0886 with isotonic calibration (both from 5-fold cross-validated probabilities) — calibration made no measurable difference, so `CalibratedClassifierCV` was not adopted for the final pipeline. HistGradientBoosting's probabilities were already well-calibrated out of the box.

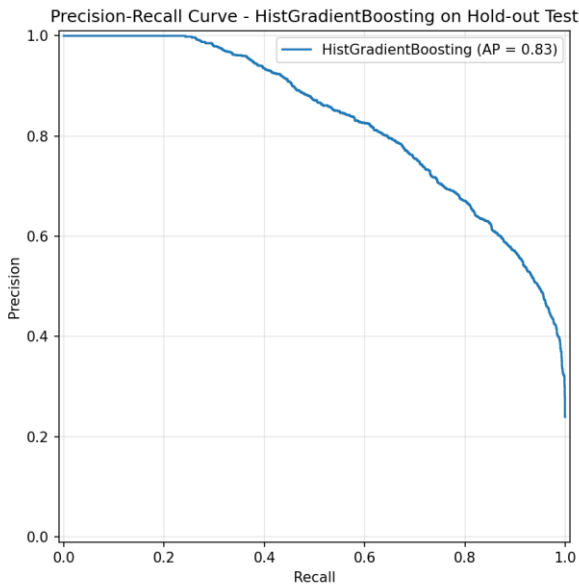
Threshold tuning to maximize F1 (on cross-validated training probabilities) found threshold = 0.40 beats the default 0.5: F1 rises from 0.710 to 0.725 by trading some precision for recall (fewer missed true positives). This threshold is used for the final saved artifact.

TIP: If the business objective shifts toward minimizing false positives (e.g. cost of wrong outreach is high), a higher threshold such as 0.70 (precision $\approx$ 0.90, recall $\approx$ 0.45) is available from the same probability output — no retraining needed.

4. Final Evaluation (Hold-Out Test Set)



Threshold	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
0.50 (default)	0.877	0.792	0.660	0.720	0.930	0.834
0.40 (F1-tuned)	0.870	0.724	0.740	0.732	0.930	0.834



The tuned threshold improves F1 and recall on the untouched hold-out test set, consistent with what the cross-validated training predictions predicted — a good sign the threshold choice generalizes rather than being overfit to the training folds.

5. Saved Artifact & Reproducibility

- Artifact: `artifacts/final_pipeline.joblib` — contains the full fitted pipeline (feature engineering + preprocessing + tuned HistGradientBoosting), the chosen threshold (0.40), best hyperparameters, and library versions (scikit-learn 1.8.0, numpy, pandas).
- All randomness is seeded (`random_state=42`) across the split, CV folds, and every estimator, so re-running the script reproduces these results.
- Expected production behavior: performance should track these hold-out numbers as long as new data resembles this training distribution; unseen categorical values are handled safely (encoded as zero) but silently lose signal, and population drift (e.g. a shifting income distribution) would call for periodic re-evaluation.